

Lab 9: What the Country Was Looking At

84-355 Democracy's Data | Week 9

Your name here

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Before we start

Chapter 8 asks what the media does to voters. That question is hard to answer because the two things you would need — what people were exposed to, and what they came to believe — are both very difficult to measure.

Today we measure something narrower and much more tractable: **what people went looking for**. Every time someone reads an English Wikipedia article, that request is counted. The counts are public, daily, and go back years.

This is not exposure and it is certainly not persuasion. It is *curiosity*, and curiosity leaves a sharper trace than you might expect. By the end of this lab you will be able to date the turning points of the 2024 campaign to within a day without looking at a single news story.

Everything is base R. Work in order.

Part 1: A year of attention

```
w <- read.csv("data/wiki_attention_2024.csv", stringsAsFactors = FALSE)
ev <- read.csv("data/campaign_events_2024.csv", stringsAsFactors = FALSE)
w$date <- as.Date(w$date)
ev$date <- as.Date(ev$date)
dim(w)
```

```
#> [1] 4391    3
```

Twelve articles, every day of 2024, in **long format** — one row per article per day.

```
head(w, 3)
```

```
#>      date                                article views
#> 1 2024-01-01 2024_United_States_presidential_election 39795
#> 2 2024-01-02 2024_United_States_presidential_election 41478
#> 3 2024-01-03 2024_United_States_presidential_election 39879
```

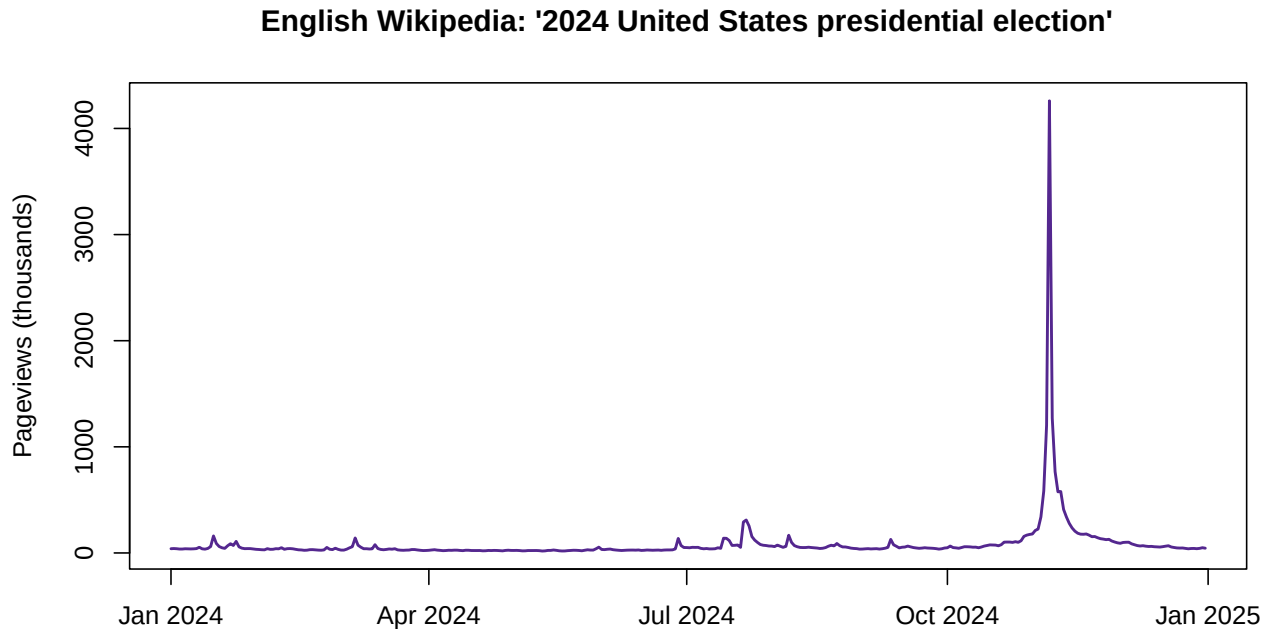
```
table(w$article)
```

```
#>
#> 2024_United_States_presidential_election
#>                                     366
#>                                Donald_Trump
#>                                     366
#> Electoral_College_(United_States)
#>                                     366
#>                                Haitian_Americans
#>                                     366
#>                                JD_Vance
#>                                     365
#>                                Joe_Biden
#>                                     366
#>                                Kamala_Harris
#>                                     366
#>                                Opinion_poll
#>                                     366
#>                                Project_2025
#>                                     366
#>                                Springfield,_Ohio
#>                                     366
#>                                Swing_state
#>                                     366
#>                                Tim_Walz
#>                                     366
```

Part 2: Attention is spiky, not steady

Start with the article for the election itself.

```
e <- w[w$article == "2024_United_States_presidential_election", ]  
  
plot(e$date, e$views / 1000, type = "l", lwd = 1.8, col = "#54278F",  
      xlab = "", ylab = "Pageviews (thousands)",  
      main = "English Wikipedia: '2024 United States presidential election'")
```



```
c(median = median(e$views), peak = max(e$views),  
  ratio = round(max(e$views) / median(e$views)))
```

```
#> median    peak    ratio  
#>  43122 4260085      99
```

A typical day drew about **43,000** readers. The peak drew **4,260,085** — **ninety-nine times** the median. Almost everything that happens in public attention happens on a handful of days.

That is the single most important fact about the media environment, and it is visible before we have named a single event.

Part 3: The day the race changed

Now the two principals, plus Biden.

```

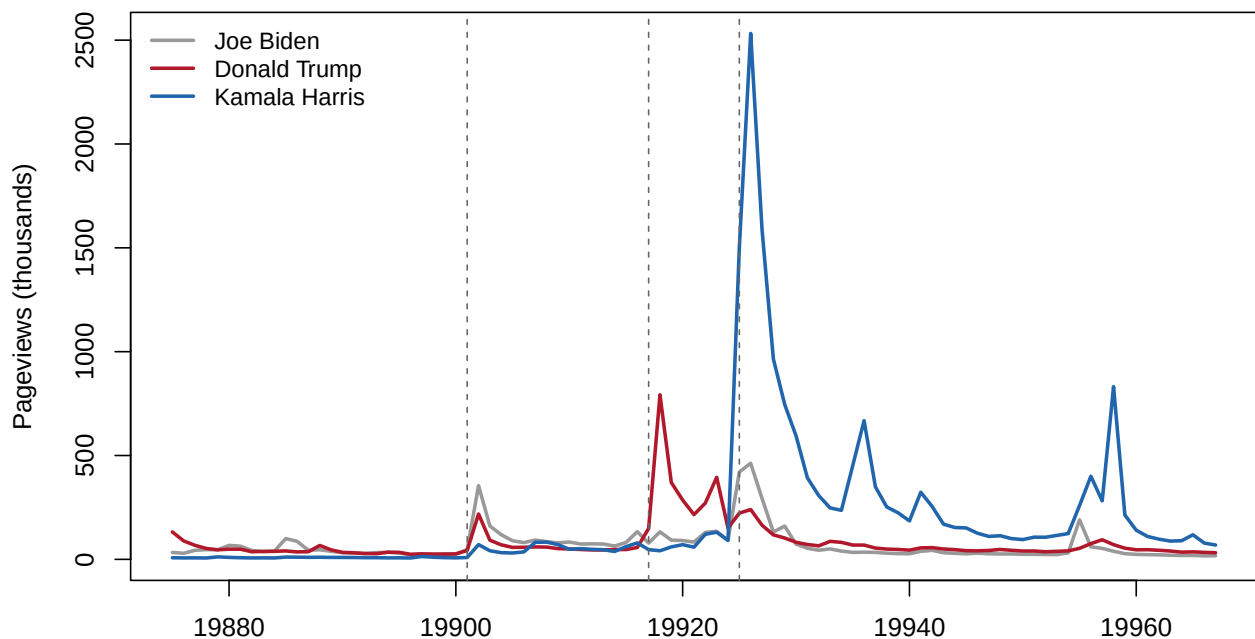
cands <- c("Joe_Biden", "Donald_Trump", "Kamala_Harris")
cols <- c("#999999", "#B2182B", "#2166AC")

sub <- w[w$article %in% cands & w$date >= as.Date("2024-06-01") &
      w$date <= as.Date("2024-09-01"), ]

plot(NULL, xlim = range(sub$date), ylim = c(0, max(sub$views) / 1000),
      xlab = "", ylab = "Pageviews (thousands)",
      main = "Summer 2024")
for (i in seq_along(cands)) {
  s <- sub[sub$article == cands[i], ]
  lines(s$date, s$views / 1000, lwd = 2.2, col = cols[i])
}
abline(v = as.Date(c("2024-06-27", "2024-07-13", "2024-07-21")),
       lty = 2, col = "grey40")
legend("topleft", gsub("_", " ", cands), col = cols, lwd = 2.2,
       bty = "n", cex = 0.9)

```

Summer 2024



Three dashed lines: the June debate, the assassination attempt, and Biden's withdrawal. Look at what each one does.

```

h <- w[w$article == "Kamala_Harris", ]
h[h$date >= as.Date("2024-07-19") & h$date <= as.Date("2024-07-24"), ]

```

```

#>      date      article  views
#> 2396 2024-07-19 Kamala_Harris 130796
#> 2397 2024-07-20 Kamala_Harris  91314
#> 2398 2024-07-21 Kamala_Harris 1501857
#> 2399 2024-07-22 Kamala_Harris 2532334
#> 2400 2024-07-23 Kamala_Harris 1588200
#> 2401 2024-07-24 Kamala_Harris  964676

```

```
round(max(h$views) / median(h$views))
```

```
#> [1] 204
```

Harris drew about **12,400** readers on a typical day of 2024. On **22 July**, the day after Biden withdrew, she drew **2,532,334** — roughly **200 times** her own baseline.

Two and a half million people went to look up who she was.

Check the events file rather than trusting the eye:

```
ev
```

```
#>      date                                event
#> 1 2024-06-27      First presidential debate (Biden-Trump)
#> 2 2024-07-13      Assassination attempt on Trump, Butler PA
#> 3 2024-07-15      Vance named Republican VP nominee
#> 4 2024-07-21      Biden withdraws, endorses Harris
#> 5 2024-08-06      Walz named Democratic VP nominee
#> 6 2024-08-19      Democratic National Convention opens
#> 7 2024-09-10 Harris-Trump debate; false Springfield claim aired
#> 8 2024-10-01      Vance-Walz vice-presidential debate
#> 9 2024-11-05      Election Day
```

Part 4: From nobody to nominee in a day

The running mates are the clearest case in the whole dataset, because before their selection almost nobody was looking for them at all.

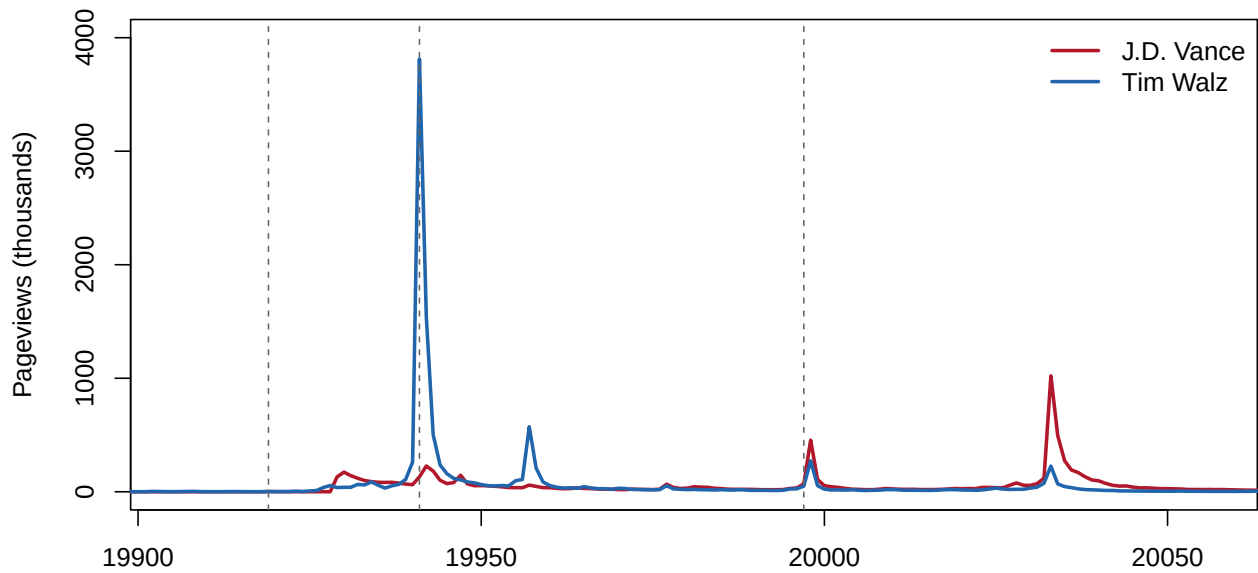
```
vp <- w[w$article %in% c("JD_Vance", "Tim_Walz"), ]
aggregate(views ~ article, data = vp, FUN = median)
```

```
#>   article  views
#> 1 JD_Vance   23.0
#> 2 Tim_Walz 1242.5
```

The median day of 2024 sent 23 people to J.D. Vance's Wikipedia article. Twenty-three. He was a sitting United States senator.

```
plot(NULL, xlim = as.Date(c("2024-07-01", "2024-11-30")),
     ylim = c(0, 4000), xlab = "", ylab = "Pageviews (thousands)",
     main = "Two running mates")
for (i in seq_along(c("JD_Vance", "Tim_Walz"))) {
  a <- c("JD_Vance", "Tim_Walz")[i]
  s <- vp[vp$article == a, ]
  lines(s$date, s$views / 1000, lwd = 2.2, col = c("#B2182B", "#2166AC")[i])
}
abline(v = as.Date(c("2024-07-15", "2024-08-06", "2024-10-01")), lty = 2,
      col = "grey40")
legend("topright", c("J.D. Vance", "Tim Walz"), col = c("#B2182B", "#2166AC"),
      lwd = 2.2, bty = "n")
```

Two running mates



```
for (a in c("JD_Vance", "Tim_Walz")) {
  s <- vp[vp$article == a, ]
  cat(a, "- peak", format(max(s$views), big.mark = ","),
      "on", format(s$date[which.max(s$views)]), "\n")
}
```

```
#> JD_Vance - peak 1,021,207 on 2024-11-06
#> Tim_Walz - peak 3,811,244 on 2024-08-06
```

Walz's announcement day drew **3,811,244** readers, against a median day of 1,242.5 across the year. Vance's own peak is printed above too.

Hold on to both numbers. We come back to them in Part 6.

Part 5: Misinformation leaves a footprint

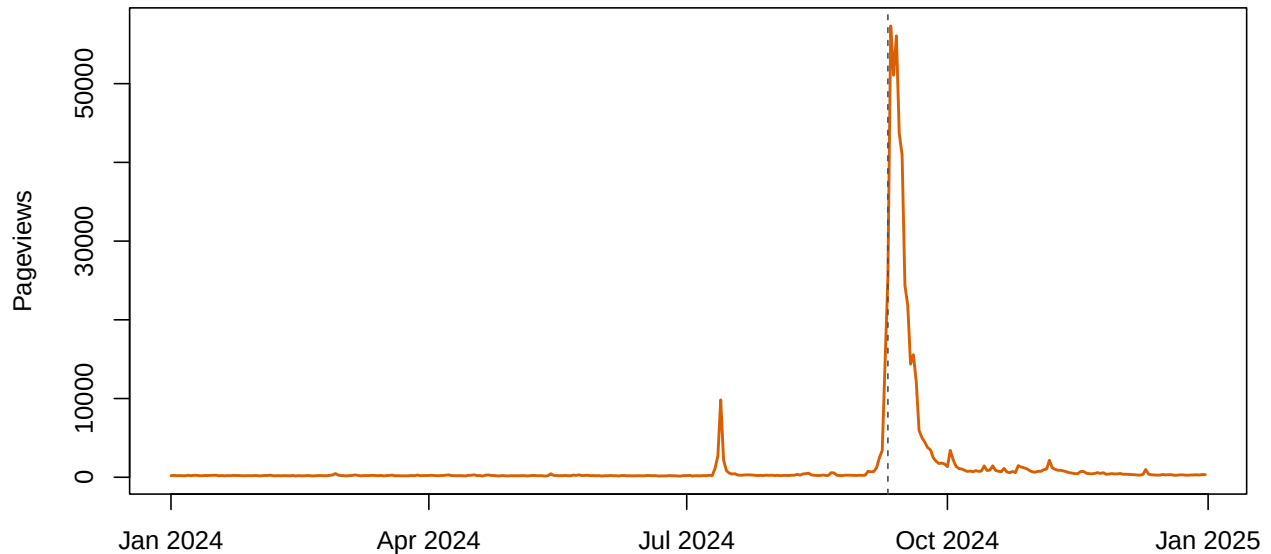
On 10 September 2024, during the presidential debate, a false claim was made that Haitian immigrants in Springfield, Ohio were eating their neighbours' pets. The claim was investigated by local officials and found to be baseless.

The claim was false. The attention was not.

```
sp <- w[w$article == "Springfield,_Ohio", ]

plot(sp$date, sp$views, type = "l", lwd = 1.8, col = "#D95F02",
     xlab = "", ylab = "Pageviews",
     main = "English Wikipedia: 'Springfield, Ohio'")
abline(v = as.Date("2024-09-10"), lty = 2, col = "grey30")
```

English Wikipedia: 'Springfield, Ohio'



```
c(median_day = median(sp$views),  
  peak       = max(sp$views),  
  peak_date  = format(sp$date[which.max(sp$views)]))
```

```
#> median_day      peak    peak_date  
#>      "242"      "57339" "2024-09-11"
```

```
round(max(sp$views) / median(sp$views))
```

```
#> [1] 237
```

A town of 60,000 people drew **242** readers on a normal day and **57,339** the day after the debate — **237 times** its baseline.

The effect is not confined to the town:

```
ha <- w[w$article == "Haitian_Americans", ]  
c(median_day = median(ha$views), peak = max(ha$views),  
  peak_date  = format(ha$date[which.max(ha$views)]))
```

```
#> median_day      peak    peak_date  
#>      "174"      "3984" "2024-09-13"
```

This is what Chapter 8 means by an agenda-setting effect, caught in the act. The media did not tell these hundreds of thousands of people what to think about Springfield. It told them that Springfield was a thing to think about at all — and a false statement did that just as efficiently as a true one would have.

Part 6: What attention is not

Three cautions, and the third is the one that matters.

Pageviews are not exposure. Far more people saw a headline about Springfield than looked it up. This measures the fraction curious enough to act.

Pageviews are not belief. Someone looking up Springfield may have been checking whether the claim was true, appalled by it, or looking for an address. The number cannot tell them apart.

And attention is not support.

Before you run the next chunk, write a ranking down. Four people stood on the two national tickets in 2024: Harris, Trump, Vance, Walz. Put all four in order by each one's **single largest day** of readership — not the total for the year, not how the election came out, just each person's one biggest day. **Who is first, and who is fourth?** Three of those four numbers have already gone past you in this lab. Commit to the whole order of four, on paper, before you run the chunk.

```
peaks <- aggregate(views ~ article, data = w[w$article %in%  
  c("Donald_Trump", "Kamala_Harris", "JD_Vance", "Tim_Walz"), ], FUN = max)  
peaks[order(-peaks$views), ]
```

```
#>      article  views  
#> 4      Tim_Walz 3811244  
#> 1  Donald_Trump 2782082  
#> 3  Kamala_Harris 2532334  
#> 2      JD_Vance 1021207
```

Tim Walz has the largest single-day attention figure of any candidate in 2024 — larger than Trump's peak the day after the election. His ticket lost.

Now check your ranking. Hardly anyone puts Walz first and Vance fourth, because the order most people have in mind is an order of fame, or of how the election turned out. This table is neither. It ranks the four by how many strangers, on one day, needed to find out who they were — and three of the four numbers had already gone past you before you guessed.

If attention converted into votes, that table would be a forecast. It is not one. It is a record of what surprised people — and novelty is not popularity. A country looking something up is a country that does not already know.

Part 7: On your own

This is the part you turn in. **You are not being asked to write code.** Each option is a working chunk with one thing to change, or nothing at all.

Pick **one**.

Option A — Pick an article and date its spikes


```

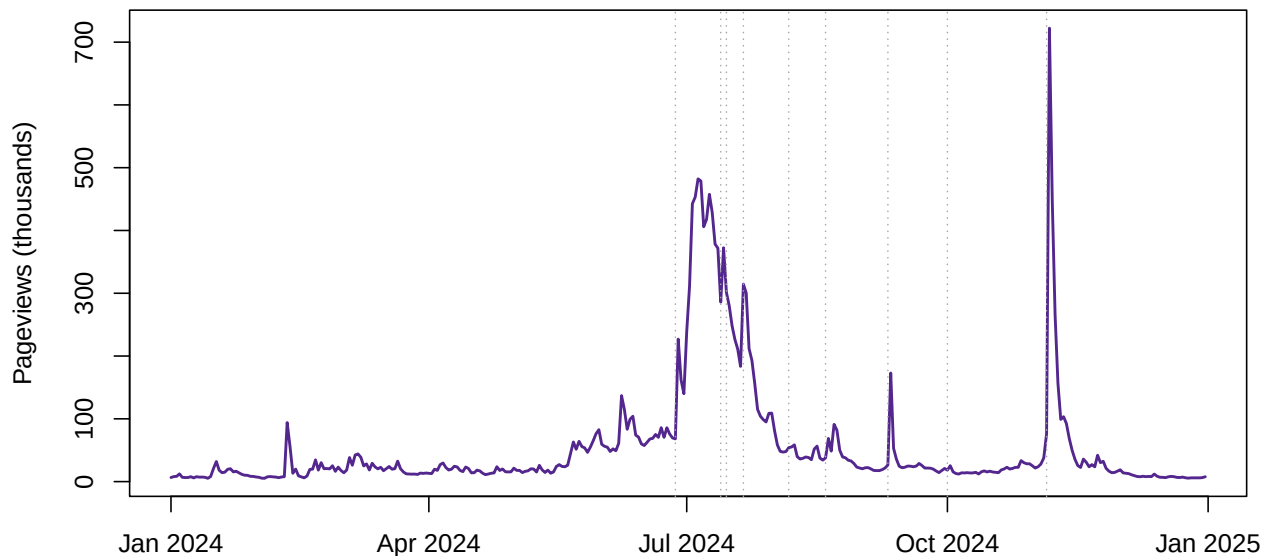
this_article <- "Project_2025"      # <- change to any article in the data

s <- w[w$article == this_article, ]

plot(s$date, s$views / 1000, type = "l", lwd = 1.8, col = "#54278F",
      xlab = "", ylab = "Pageviews (thousands)",
      main = gsub("_", " ", this_article))
abline(v = ev$date, lty = 3, col = "grey65")

```

Project 2025



```

top <- s[order(-s$views), ][1:6, ]
top[order(top$date), ]

```

```

#>      date      article  views
#> 3113 2024-07-04 Project_2025 453288
#> 3114 2024-07-05 Project_2025 482218
#> 3115 2024-07-06 Project_2025 478842
#> 3118 2024-07-09 Project_2025 457590
#> 3238 2024-11-06 Project_2025 722256
#> 3239 2024-11-07 Project_2025 448452

```

```

c(median = median(s$views), peak = max(s$views),
  ratio = round(max(s$views) / median(s$views)))

```

```

#>  median    peak    ratio
#> 21690.5 722256.0    33.0

```

Write about: name the days. Do the biggest spikes line up with the events in `ev`, or with something the events file does not contain? Project 2025 is interesting precisely because its spikes are *not* all set-piece campaign moments — some are the product of one side deliberately drawing attention to the other side’s document. What does it mean for “media effects” when a campaign succeeds in making people look something up?

Try `Swing_state` too, and say what its shape tells you about when ordinary people start paying attention to the mechanics of an election.

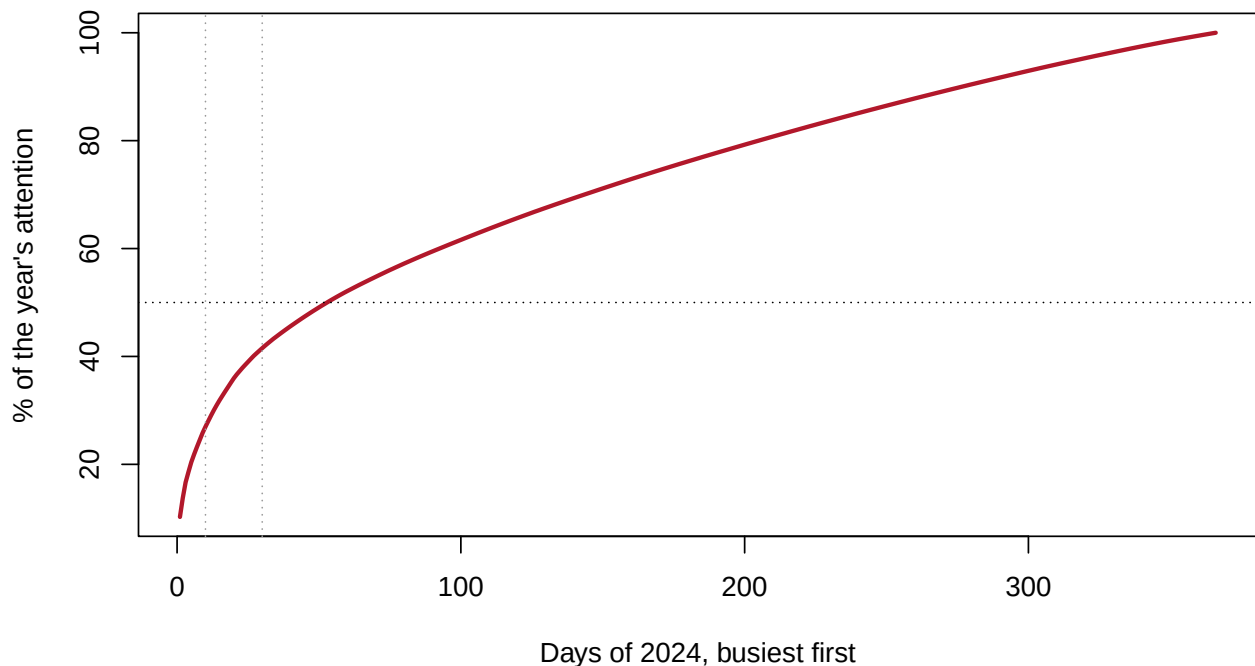
Option B — How concentrated is attention?

```
this_article <- "Donald_Trump"      # <- change this

s <- w[w$article == this_article, ]
s <- s[order(-s$views), ]
s$cum <- cumsum(s$views) / sum(s$views) * 100

plot(seq_along(s$cum), s$cum, type = "l", lwd = 2.5, col = "#B2182B",
     xlab = "Days of 2024, busiest first", ylab = "% of the year's attention",
     main = paste("Concentration of attention:", gsub("_", " ", this_article)))
abline(h = 50, lty = 3); abline(v = c(10, 30), lty = 3, col = "grey60")
```

Concentration of attention: Donald Trump



```
c(top_10_days = round(s$cum[10], 1),
  top_30_days = round(s$cum[30], 1),
  days_for_half = which(s$cum >= 50)[1])
```

```
#>   top_10_days  top_30_days days_for_half
#>         27.0         41.5         53.0
```

Write about: how many days of the year account for half of all the attention this article received? Compare a principal (Trump, Harris) against a topic (Opinion_poll, Electoral_College_(United_States)). Which

is spikier, and why would that be? Then the question worth arguing: if public attention to politics is compressed into a handful of days, what does that imply about a campaign event's value — and about the strategy of *creating* such days?

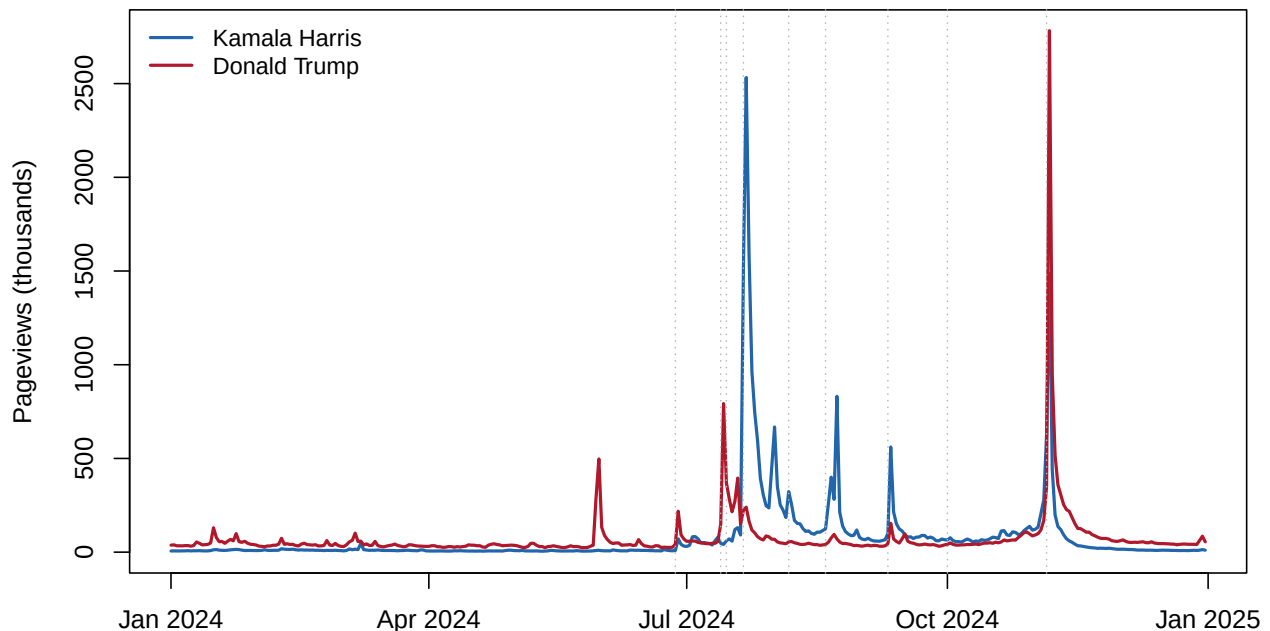
Option C — Two candidates, side by side

```
a1 <- "Kamala_Harris"           # <- change either of these
a2 <- "Donald_Trump"

s1 <- w[w$article == a1, ]; s2 <- w[w$article == a2, ]

plot(s1$date, s1$views / 1000, type = "l", lwd = 2, col = "#2166AC",
     ylim = c(0, max(c(s1$views, s2$views)) / 1000),
     xlab = "", ylab = "Pageviews (thousands)",
     main = paste(gsub("_", " ", a1), "vs", gsub("_", " ", a2)))
lines(s2$date, s2$views / 1000, lwd = 2, col = "#B2182B")
abline(v = ev$date, lty = 3, col = "grey70")
legend("topleft", gsub("_", " ", c(a1, a2)), col = c("#2166AC", "#B2182B"),
      lwd = 2, bty = "n", cex = 0.9)
```

Kamala Harris vs Donald Trump



```
# Who led on how many days, before and after Biden withdrew?
mg <- merge(s1[, c("date", "views")], s2[, c("date", "views")], by = "date",
           suffixes = c("_1", "_2"))
mg$phase <- ifelse(mg$date < as.Date("2024-07-21"), "before Jul 21", "after Jul 21")
table(mg$phase, ifelse(mg$views_1 > mg$views_2, a1, a2))
```

```
#>
#>           Donald_Trump Kamala_Harris
#>   after Jul 21           57           107
#>   before Jul 21          194            8
```

Write about: who “won” the attention battle, and does the answer change depending on which half of the year you look at? Be careful with the word won — say explicitly what you think leading in pageviews does and does not indicate. Then connect it to Part 6: Harris led on many days after July and lost the election. Is attention worth anything at all to a campaign, and if so, what?

Option D — Ask your own question

Any article in the file is fair game. Compare a person to an institution. Look for a day with a spike nobody has explained yet. Follow whatever nags.

What to submit. Knit to HTML or PDF and upload to Canvas, with your chosen chunk and **two or three paragraphs** of interpretation.

The writing is the assignment. I am not grading your code — it is my code. I am grading whether you can look at a curve of public attention and say something careful about what it does and does not show.

Where the data came from

Wikimedia Foundation Pageviews API (https://wikimedia.org/api/rest_v1/#/Pageviews%20data), daily counts for English Wikipedia, 1 January to 31 December 2024. Wikimedia publishes these counts under CC0, which is why this lab can ship its own copy of them.

Two choices in how the data was pulled, both of which matter:

- **Human traffic only.** The API can return all requests including known bots and crawlers. This file uses the `user` filter, which excludes them. Raw all-agent counts are badly contaminated and would show spikes that no person ever caused.
- **English Wikipedia only.** A large share of the world’s interest in an American election arrives in other languages, and none of it is here.

The deeper limit is the one Part 6 is about. A pageview is a person choosing to look something up. It is a real behaviour, publicly and consistently measured, and it is *not* the thing Chapter 8 is ultimately asking about. Nothing in this file tells you what anyone believed before or afterwards.

Campaign events in `campaign_events_2024.csv` were compiled by hand from the public record and are deliberately kept separate from the pageview file — they are the independent yardstick the spikes get checked against, not something derived from the spikes themselves.

The script that fetched, checked and wrote all of this is `data/build-data.R`.